

Analyzing the Impact of Lead Exposure on Children’s Neurological and Psychological Outcomes with Missing Data

Ziheng Zhang

1 Introduction

A cohort of children who lived near a lead smelter in El Paso and Texas was studied to investigate whether exposure to lead affects neurological and psychological functions. The participants’ ages ranged from 45 to 191 months at their first blood lead measurement in 1972 (*lead72*), and they all lived within 4.1 miles of the smelter for at least 12 of the previous 24 months. The *lead72* measurements vary between 1 and 68, with a median of 33.

The dataset includes two groups based on blood lead levels: a high-exposure group (blood lead levels ≥ 40 $\mu\text{g/ml}$) and a low-exposure comparison group (blood lead levels < 40 $\mu\text{g/ml}$). Several outcome measures were collected, including the number of finger-wrist taps (*fwtap*), two-plate taps (*tap*), auditory (*audreact*) and visual (*visreact*) reaction times, and IQ scores (*iqf*). Additionally, parent-reported variables, such as symptoms related to hyperactivity (*hypscore*), were also collected. The *hypscore* is an ordinal variable representing hyperactivity levels, ranging from 0 to 4. However, there are some missing observations, including blood lead levels for siblings of high-exposure children, which may create some challenges in the analysis.

In this report, we will first explore the missing data through visualization. Then, we will handle the data by using two different imputation methods, specifically joint-modeling imputation and fully-conditional-specification imputation, to investigate whether there are differences in neurological and psychological outcomes between two lead exposure groups and compare the results of the two methods. Finally, we will discuss a strategy for obtaining a 95% interval estimate for the average 1972 blood

lead level.

2 Exploratory Data Analysis

Missing data patterns in the dataset are shown in Figure 1. Most missing values appear in the variables *lead72*, *hypscore*, and some neurological measures. We can see that the hyperactivity score (*hypscore*) variable has the highest missing rate (32%) while blood lead measurement in 1972 (*lead72*) holds the lowest (2%). Overall, 6.1% of the data is missing.

Figure 2 shows 6 variables with missing values and interactions between these variables. All missing values in the four neurological and psychological outcomes occur simultaneously, which indicates that the missingness in these variables may be due to some systematic rather than random factors. The missing values in the hyperactivity score are mostly independent of other variables, with only 2 out of 40 cases overlapping with the missing blood lead measurement in 1972.

3 Missing Data Imputation and Analysis

3.1 Task 1: Joint-Modeling Imputation

We used the *jomo* package in R for imputation. First, we compared the results of 5 imputations with the original data in Figure 3a. The distributions of the imputed and original data are almost identical, which means our imputed values are within a plausible range.

To compare neurological and psychological differences between the two lead exposure groups, we selected *fwtap*, *tap*, and *hypscore* as outcome variables. We used the first imputed dataset for visualization. As shown in Figure 4a–Figure 6a, the results for both groups are almost the same. Figure 4a and Figure 5a show the high-

lead-exposure group has slightly higher variability of two-plate taps values and tends to have fewer finger-wrist taps and two-plate taps overall while Figure 6a illustrates low-lead-exposure group has more frequent hyperactivity and since the category “Always” does not appear in the observed data, it is theoretically impossible to appear in the imputed values, resulting in only four categories shown in the figure.

Next, we conducted t-tests for *fwtap* and *tap*, and a Wilcoxon test for *hypscore* to test whether there are significant differences between the two exposure groups. Table 1 shows that the number of finger-wrist taps differs significantly between the two groups (p-value = 0.022), which means there is a neurological difference from a marginal perspective.

In addition to the indicator variable for the lead exposure group, we included age (*age_mo*), sex (*female*), and indicators of childhood distress (*colic* and *convulse*) to fit the regression models. We fitted linear regression models for *fwtap* and *tap*, and an ordered logistic regression model for *hypscore*. Table 2–Table 4 summarize the results of regression models. The results of all three models indicate that there is no significant neurological and psychological difference between two exposure groups (p-value = 0.116, = 0.826, and = 0.240, respectively).

3.2 Task 2: Fully-Conditional-Specification Imputation

We applied the *mice* package in R for imputation. Here, we also compared the imputed data with the original one in Figure 3b in which the distributions are almost identical again.

We still used the first imputed dataset for visualization. As shown in Figure 4b–Figure 6b, the results for both groups are almost the same, and the patterns between the groups are similar to those from Section 3.1.

Hypothesis test results and regression model results are shown in Table 1 and Table 2–Table 4, respectively. We can see that there is also no significant neurological

and psychological difference between the two exposure groups (p-value = 0.017, = 0.098, = 0.656, and = 0.422, respectively).

3.3 Comparison between JM and FCS

From the results in Section 3.1 and Section 3.2, we find that almost all outcomes are consistent, including visualization, hypothesis tests, and regression models. From a marginal perspective, there are significant neurological differences between the low-lead-exposure and high-lead-exposure groups. However, when additional covariates are introduced, the neurological and psychological differences become insignificant.

4 95% Interval Estimation

To obtain a 95% interval estimate for the average 1972 blood lead level among children in the catchment area, we propose using multiple imputation methods to handle missing data, particularly addressing the missing blood lead measurements for three previously unascertained siblings. After checking the group assignment of the siblings of these missing values, we notice that these siblings are in the high-exposure group; therefore, we assume these missing values are exchangeable with other children in the same high-exposure group instead of with all sampled children. This assumption is reasonable to some extent if we consider these children’s shared living conditions and similar environmental exposures within households, which typically give rise to correlated blood lead levels among siblings.

Considering this correlation, a fully-conditional-specification imputation method incorporating family-level random effects can be employed to impute these missing values. Furthermore, data from the Census Bureau in 1972 indicates that the average number of children in a household with at least one child under age 18 is 2.22. This information further supports our assumption about family size and sibling similarity

because this data provides real-world evidence that households with children often have more than one child. So, children in the same household tend to share similar environments and exposures.

After imputing the missing data by our proposed method, the mean blood lead level and its corresponding 95% confidence interval can be calculated using standard statistical methods combined with the rules for calculating estimates from multiple imputation, including the average within-imputation variance and between-imputation variance, to properly consider imputation uncertainty.

5 Conclusion

In this report, we explored the neurological and psychological differences between the high-lead-exposure and low-lead-exposure groups using two multiple imputation methods. Our analyses indicate that, marginally, there are significant neurological differences in finger-wrist taps between the two exposure groups. However, these differences become non-significant after adjusting for age, sex, and indicators of childhood distress. Two imputation methods have similar conclusions, which demonstrates the reliability of our results.

Appendix

Supplemental Tables

Table 1: Comparison of p-values of hypothesis tests between joint-modeling and fully-conditional-specification imputation.

JM Imputation		FCS Imputation	
Test	p-value	Test	p-value
<i>t</i> -test (fwtap)	0.022	<i>t</i> -test (fwtap)	0.017
<i>t</i> -test (tap)	0.575	<i>t</i> -test (tap)	0.799
Wilcoxon (hypscore)	0.513	Wilcoxon (hypscore)	0.485

Table 2: Comparison of linear regression results of *fwtap* between joint-modeling and fully-conditional-specification imputations.

Joint-Modeling Imputation			
Term	Estimate	SE	p-value
(Intercept)	33.437	2.722	< 0.001
Age (months)	0.174	0.022	< 0.001
colic (Ref: No)	-2.529	2.365	0.344
convulse (Ref: No)	3.149	4.255	0.462
female (Ref: Male)	-0.429	1.899	0.555
Lead Exposure (Ref: Low)	-3.516	1.934	0.116

Fully-Conditional-Specification Imputation			
Term	Estimate	SE	p-value
(Intercept)	28.748	4.905	0.001
Age (months)	0.208	0.034	< 0.001
colic (Ref: No)	-2.153	2.803	0.448
convulse (Ref: No)	3.798	4.282	0.377
female (Ref: Male)	-0.726	2.053	0.725
Lead Exposure (Ref: Low)	-3.583	2.128	0.098

Table 3: Comparison of linear regression results of *tap* between joint-modeling and fully-conditional-specification imputations.

Joint-Modeling Imputation			
Term	Estimate	SE	p-value
(Intercept)	12.217	1.432	< 0.001
Age (months)	0.041	0.012	0.016
colic (Ref: No)	-0.572	1.244	0.673
convulse (Ref: No)	-1.112	2.238	0.623
female (Ref: Male)	-0.740	0.999	0.487
Lead Exposure (Ref: Low)	-0.093	1.017	0.826

Fully-Conditional-Specification Imputation			
Term	Estimate	SE	p-value
(Intercept)	10.829	1.917	< 0.001
Age (months)	0.051	0.014	0.004
colic (Ref: No)	-0.804	1.292	0.538
convulse (Ref: No)	-1.038	2.030	0.610
female (Ref: Male)	-0.758	0.931	0.417
Lead Exposure (Ref: Low)	0.426	0.954	0.656

Table 4: Comparison of ordinal regression results of *hypscore* between joint-modeling and fully-conditional-specification imputations, excluding intercepts.

Joint-Modeling Imputation			
Term	Estimate	SE	p-value
Age (months)	-0.002	0.004	0.262
colic (Ref: No)	0.662	0.464	0.081
convulse (Ref: No)	-15.726	<0.001	0.000
female (Ref: Male)	-0.183	0.367	0.288
Lead Exposure (Ref: Low)	0.220	0.381	0.240

Fully-Conditional-Specification Imputation			
Term	Estimate	SE	p-value
Age (months)	0.009	0.005	0.081
colic (Ref: No)	0.651	0.595	0.290
convulse (Ref: No)	-13.379	7.435	0.567
female (Ref: Male)	-0.349	0.444	0.440
Lead Exposure (Ref: Low)	0.334	0.413	0.422

Supplemental Figures

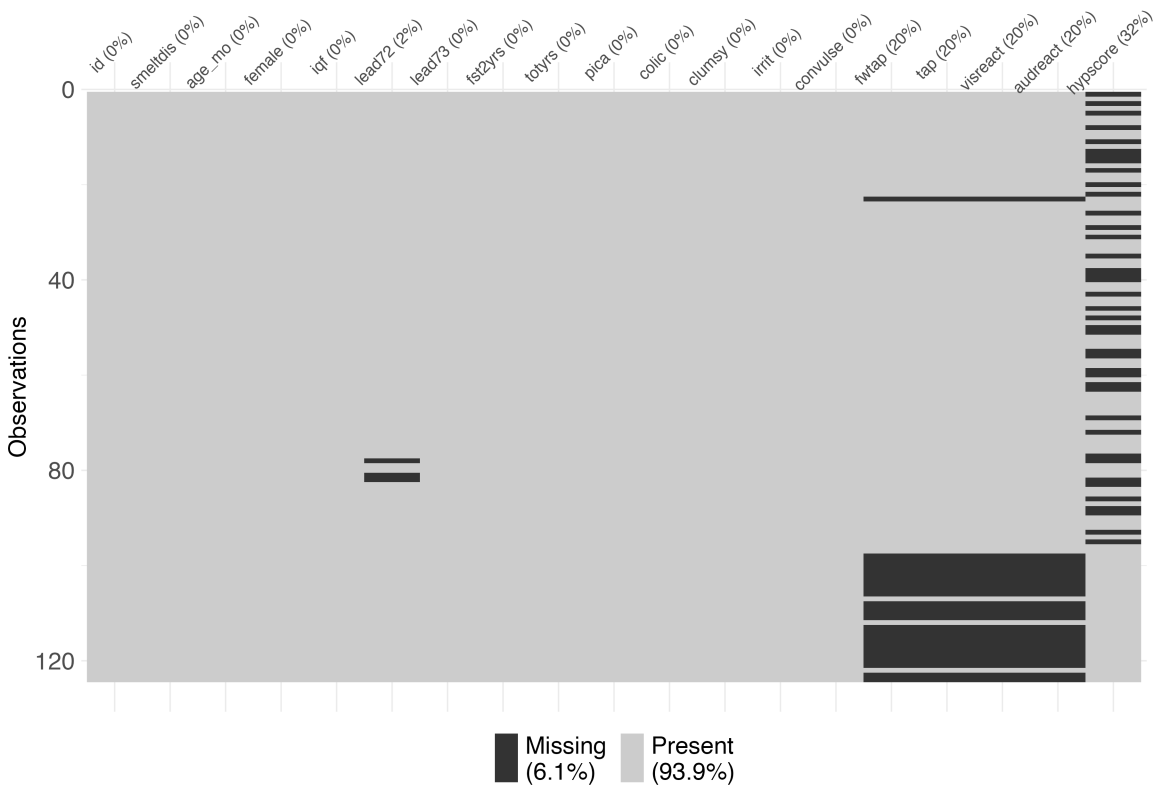


Figure 1: Missing data pattern across variables. The heatmap shows missing values (black) and observed values (gray) for each observation.

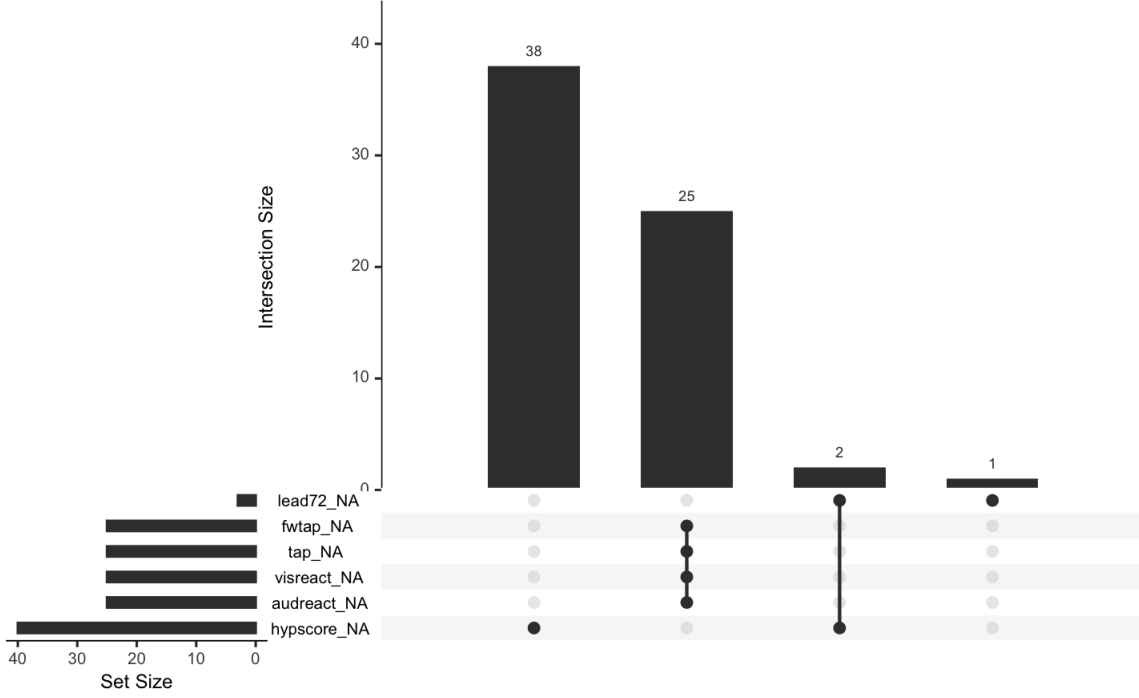
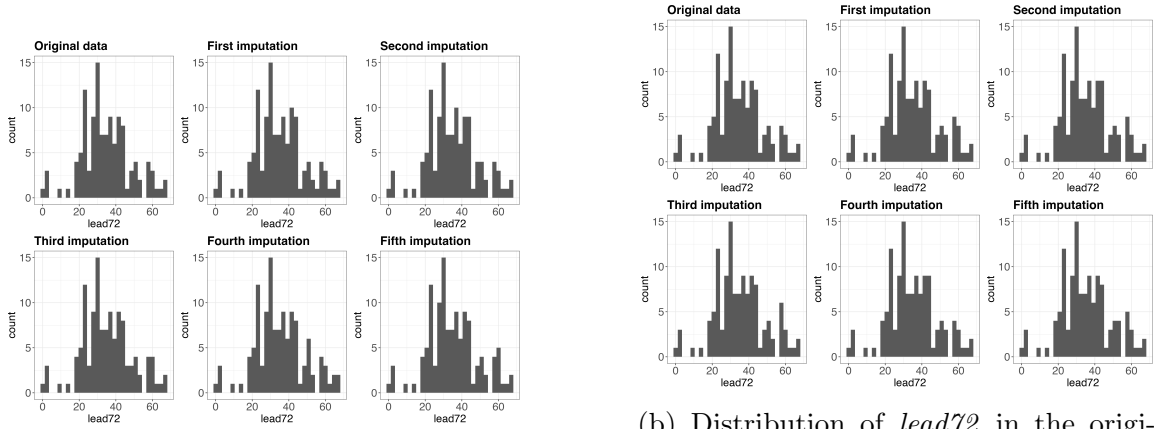


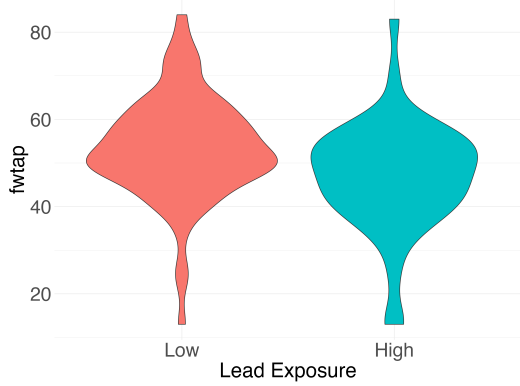
Figure 2: UpSet plot of missing data intersections. The bar chart shows the frequency of missing values in different variables. The connected dots indicate combinations of variables where missing values occur simultaneously.



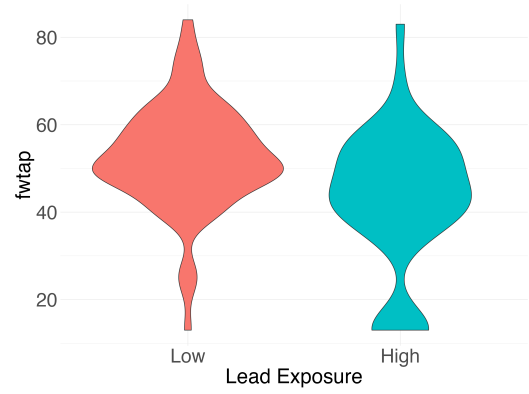
(a) Distribution of *lead72* in the original dataset and five imputed datasets using joint-modeling imputation.

(b) Distribution of *lead72* in the original dataset and five imputed datasets using fully-conditional-specification imputation.

Figure 3: Comparison of *lead72* distributions between joint-modeling and fully-conditional-specification imputations.

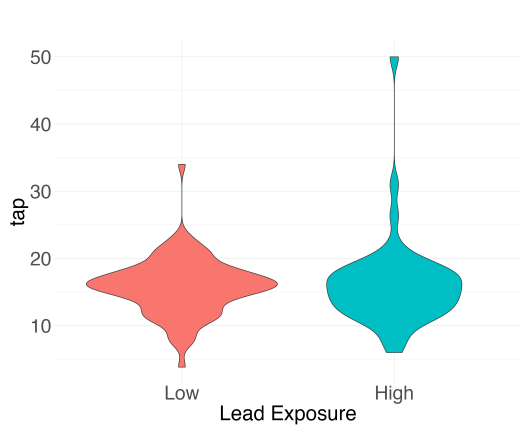


(a) Distribution of $fwtap$ by lead exposure group using joint-modeling imputation.

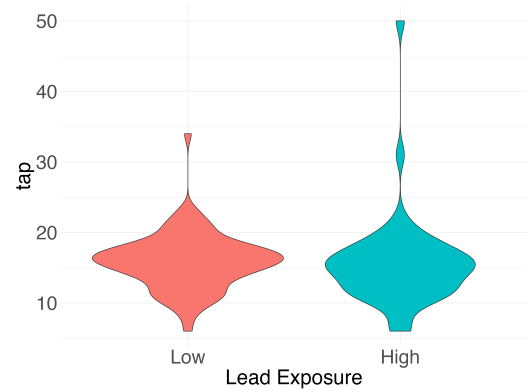


(b) Distribution of $fwtap$ by lead exposure group using fully-conditional-specification imputation.

Figure 4: Comparison of $fwtap$ distributions between joint-modeling and fully-conditional-specification imputations.

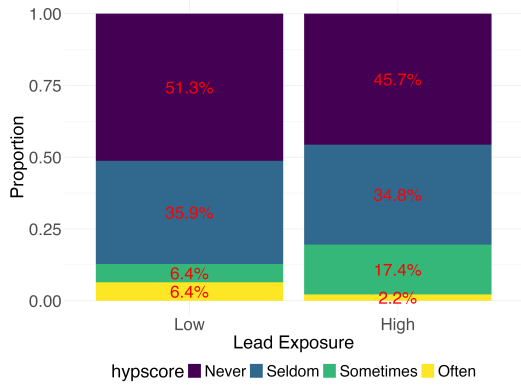


(a) Distribution of tap by lead exposure group using joint-modeling imputation.

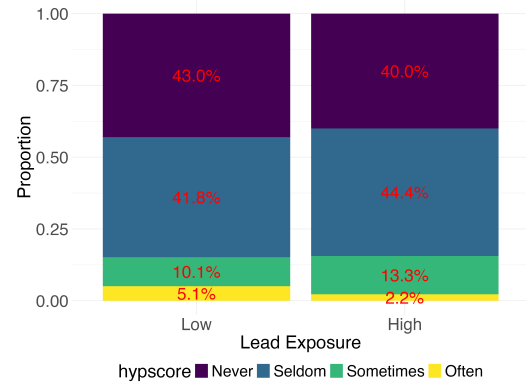


(b) Distribution of tap by lead exposure group using fully-conditional-specification imputation.

Figure 5: Comparison of tap distributions between joint-modeling and fully-conditional-specification imputations.



(a) Proportion of *hypscore* by lead exposure group using joint-modeling imputation.



(b) Proportion of *hypscore* by lead exposure group using fully-conditional-specification imputation.

Figure 6: Comparison of *hypscore* proportions between joint-modeling and fully-conditional-specification imputations.

Supplemental Code

Code for Joint-Modeling Imputation

```
lead <- read.table("lead.txt")
lead[lead == -9] <- NA
col_names <- c("id", "smeltdis", "age_mo", "female", "iqf", "lead72", "lead73",
               "fst2yrs", "totyrs", "pica", "colic", "clumsy", "irrit",
               "convulse", "fwtap", "tap", "visreact", "audreact", "hypsore")
colnames(lead) <- col_names

Y.con <- lead[, c("id", "age_mo", "iqf", "lead72", "lead73", "totyrs", "fwtap",
                 "tap", "visreact", "audreact")]
Y.cat <- lead[, c("smeltdis", "female", "fst2yrs", "pica", "colic", "clumsy",
                 "irrit", "convulse", "hypsore")]
Y.numcat <- matrix(c(3, 2, 2, 2, 2, 2, 2, 2, 4), 1, 9)

lead <- jomo1mix(Y.con, Y.cat, Y.numcat)

lead$smeltdis <- factor(lead$smeltdis, levels = c(1,2,3),
                       labels = c("0-1 miles", "1-2.5 miles", "2.5-4.1 miles"),
                       ordered = TRUE)

lead$female <- factor(lead$female, levels = c(0,1), labels = c("Male", "Female"))
lead$fst2yrs <- factor(lead$fst2yrs, levels = c(0,1), labels = c("No", "Yes"))
lead$pica <- factor(lead$pica, levels = c(0,1), labels = c("No", "Yes"))
lead$colic <- factor(lead$colic, levels = c(0,1), labels = c("No", "Yes"))
lead$clumsy <- factor(lead$clumsy, levels = c(0,1), labels = c("No", "Yes"))
lead$irrit <- factor(lead$irrit, levels = c(0,1), labels = c("No", "Yes"))
```

```

lead$convulse <- factor(lead$convulse, levels = c(0,1), labels = c("No", "Yes"))
lead$hypscore <- factor(lead$hypscore, levels = c(0,1,2,3,4),
                        labels = c("Never", "Seldom", "Sometimes", "Often",
                                   "Always"), ordered = TRUE)

lead <- lead[, -c(20:21)]

lead_mice <- lead[125:744, ] |>
  mutate(lead_exposure = ifelse(lead72 >= 40, "High", "Low"),
         lead_exposure = factor(lead_exposure, levels = c("Low", "High")))
lead_mice$Imputation <- as.factor(lead_mice$Imputation)
lead_list <- split(lead_mice, lead_mice$Imputation)

lm_results <- lapply(lead_list, function(data) {
  lm(fwtap ~ lead_exposure + colic + convulse + age_mo + female, data = data)
})
lm_summaries <- lapply(lm_results, tidy)
lm_df <- bind_rows(lm_summaries, .id = "Imputation")
fwtap_partial <- lm_df |>
  group_by(term) |>
  summarise(
    estimate = mean(estimate),
    std.error = sqrt(mean(std.error^2)),
    p.value = mean(p.value)
  )
print(fwtap_partial)

```

```

lm_results <- lapply(lead_list, function(data) {
  lm(tap ~ lead_exposure + colic + convulse + age_mo + female, data = data)
})

lm_summaries <- lapply(lm_results, tidy)
lm_df <- bind_rows(lm_summaries, .id = "Imputation")
tap_partial <- lm_df |>
  group_by(term) |>
  summarise(
    estimate = mean(estimate),
    std.error = sqrt(mean(std.error^2)),
    p.value = mean(p.value)
  )
print(tap_partial)

```

```

lm_results <- lapply(lead_list, function(data) {
  polr(hypscore ~ lead_exposure + colic + convulse + age_mo + female,
    Hess = TRUE, data = data)
})

lm_summaries <- lapply(lm_results, tidy)
lm_df <- bind_rows(lm_summaries, .id = "Imputation")
hypscore_partial <- lm_df |>
  group_by(term) |>
  summarise(
    estimate = mean(estimate),

```

```

    std.error = sqrt(mean(std.error^2)),
    p.value = mean(1-pt(abs(statistic), df = 115))
  )
print(hypscore_partial)

```

```

lm_results <- lapply(lead_list, function(data) {
  t.test(fwtap ~ lead_exposure, data = data)
})
lm_summaries <- lapply(lm_results, tidy)
lm_df <- bind_rows(lm_summaries, .id = "Imputation")
fwtap_test <- lm_df |>
  summarise(
    p.value = mean(p.value)
  )
print(fwtap_test)

```

```

lm_results <- lapply(lead_list, function(data) {
  t.test(tap ~ lead_exposure, data = data)
})
lm_summaries <- lapply(lm_results, tidy)
lm_df <- bind_rows(lm_summaries, .id = "Imputation")
tap_test <- lm_df |>
  summarise(
    p.value = mean(p.value)
  )

```

```

print(tap_test)

lm_results <- lapply(lead_list, function(data) {
  wilcox.test(as.numeric(data$hypsore) ~ data$lead_exposure)
})

lm_summaries <- lapply(lm_results, tidy)

lm_df <- bind_rows(lm_summaries, .id = "Imputation")

hypsore_test <- lm_df |>
  summarise(
    p.value = mean(p.value)
  )

print(hypsore_test)

```

Code for Fully-Conditional-Specification Imputation

```

lead <- read.table("lead.txt")

lead[lead == -9] <- NA

col_names <- c("id", "smeltdis", "age_mo", "female", "iqf", "lead72", "lead73",
  "fst2yrs", "totyrs", "pica", "colic", "clumsy", "irrit",
  "convulse", "fwtap", "tap", "visreact", "audreact", "hypsore")

colnames(lead) <- col_names

lead$smeltdis <- factor(lead$smeltdis, levels = c(1,2,3),
  labels = c("0-1 miles", "1-2.5 miles", "2.5-4.1 miles"),
  ordered = TRUE)

lead$female <- factor(lead$female, levels = c(0,1), labels = c("Male", "Female"))

```



```

lead$fst2yrs <- factor(lead$fst2yrs, levels = c(0,1), labels = c("No", "Yes"))
lead$pica <- factor(lead$pica, levels = c(0,1), labels = c("No", "Yes"))
lead$colic <- factor(lead$colic, levels = c(0,1), labels = c("No", "Yes"))
lead$clumsy <- factor(lead$clumsy, levels = c(0,1), labels = c("No", "Yes"))
lead$irrit <- factor(lead$irrit, levels = c(0,1), labels = c("No", "Yes"))
lead$convulse <- factor(lead$convulse, levels = c(0,1), labels = c("No", "Yes"))
lead$hypscore <- factor(lead$hypscore, levels = c(0,1,2,3,4),
                        labels = c("Never", "Seldom", "Sometimes", "Often",
                                   "Always"), ordered = TRUE)

lead_mice <- mice(lead, m = 5)

lead_mice_completed <- complete(lead_mice, "long", include = TRUE) |>
  mutate(lead_exposure = ifelse(lead72 >= 40, "High", "Low"),
         lead_exposure = factor(lead_exposure, levels = c("Low", "High")))
lead_mice <- as.mids(lead_mice_completed)

fwtap_partial <- with(lead_mice, lm(fwtap ~ lead_exposure + colic + convulse +
                                   age_mo + female))

summary(pool(fwtap_partial))

tap_partial <- with(lead_mice, lm(tap ~ lead_exposure + colic + convulse +
                                   age_mo + female))

summary(pool(tap_partial))

```

```

hypscore_partial <- with(lead_mice, polr(hypscore ~ lead_exposure + colic + convulse
                                         age_mo + female, Hess = TRUE))

summary(pool(hypscore_partial))

fwtap_test <- with(lead_mice, t.test(fwtap ~ lead_exposure))
mean(sapply(fwtap_test$analyses[1:5], function(x) x$p.value))

tap_test <- with(lead_mice, t.test(tap ~ lead_exposure))
mean(sapply(tap_test$analyses[1:5], function(x) x$p.value))

hypscore_test <- with(lead_mice, wilcox.test(as.numeric(hypscore) ~ lead_exposure))
mean(sapply(hypscore_test$analyses[1:5], function(x) x$p.value))

```